DATA MINING · TECHNICAL PRESENTATION

Look First, Then Measure

Exploring data honestly, and measuring how similar two things really are

Christian Marcos I-Shou University · Fall 2026

AGENDA

Two steps before any model

TOPIC 1 · WEEK 2

Exploring data

Summary statistics, charts that don’t lie, and OLAP. Worked: a job ad’s “average salary”, and a bubble tea chain’s sales cube.

TOPIC 2 · WEEK 3

Similarity & distance

Euclidean, Manhattan, scaling, Jaccard, cosine and k-nearest neighbours. Worked: delivery, banking, shopping, hiring and loans.

Then: how the two topics connect, and what to remember.

TOPIC 1 · EXPLORING DATA

Look at the data before you trust it

Exploratory data analysis (EDA) summarises a dataset's main features, often visually, **before** any modelling, to catch problems and suggest questions.

Numbers

Where is the centre? How spread out? What shape?

Pictures

Histograms, box plots, scatter plots show what a number hides.

Cubes (OLAP)

Re-slice the same totals by store, month, product.

> Real case from the course: before a telecom built a churn model, simply looking revealed a filler value in 40% of one column and two "independent" features correlated at 0.97.

WORKED EXAMPLE 1 · MEASURES OF LOCATION

“Average salary NT$81,100”: true, but misleading

A 10-person startup’s monthly salaries (NT$ thousand): **38, 40, 42, 42, 45, 47, 50, 52, 55** and the founder’s **400**.

Statistic	Calculation	Result
Mean	811 / 10	81.1
Median	(45 + 47) / 2	46
Mode	most frequent	42
10% trimmed	drop 38 and 400: 373 / 8	46.6

Job ad says

NT$81,100

A typical employee earns

NT$46,000

Nine of ten people earn less than the “average”. One extreme value pulls the **mean**; the **median** only cares about rank order.

WORKED EXAMPLE 2 · SPREAD AND OUTLIERS

Which rule catches the founder's salary?

Five-number summary & Tukey's fences

Step	Value (NT$ k)
Min · Q1 · Median · Q3 · Max	38 · 42 · 46 · 52 · 400
IQR = Q3 − Q1	10
Fences: Q1 − 1.5 IQR, Q3 + 1.5 IQR	27 to 67
Outside the fences	400: flagged

The z-score rule (|z| > 3)

Step	Value
Mean · std dev	81.1 · 112.2
z = (400 − 81.1) / 112.2	2.84
|z| > 3?	No: missed

The outlier inflates the mean and std dev it is judged against, so it hides itself. Robust rules (IQR, median, MAD = 5) are not fooled.

TOPIC 1 · SHAPE AND RELATIONSHIPS

Two numbers never show the shape

Skewness

A long tail to the right, like salaries or house prices, puts the mean above the median (81.1 vs 46).

Mean far from median? Report both.

Modality

Two peaks often mean two hidden groups, such as exam scores from two classes merged into one column.

The mean may sit where no one is.

Correlation

r runs from −1 to +1 and measures only **linear** association between two columns.

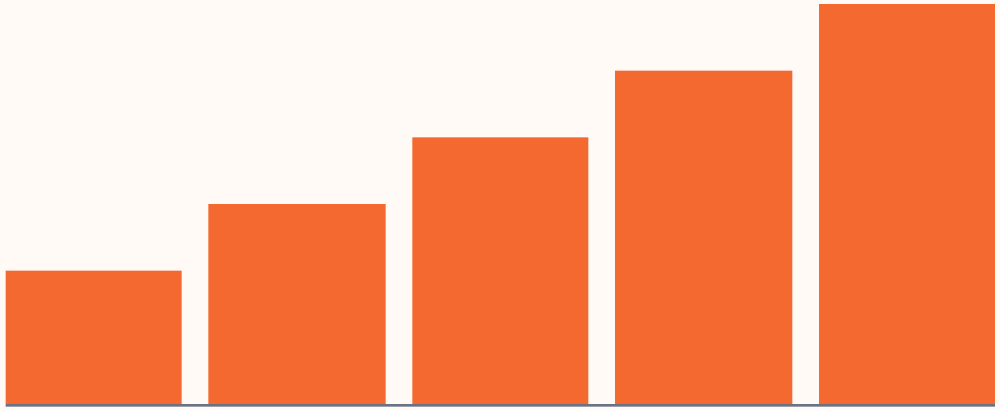
r ≈ 0 can still hide a curve.

Anscombe's quartet: four datasets with the same mean, variance and correlation (r = 0.82) look completely different when plotted. Always plot.

WORKED EXAMPLE 3 · VISUALIZATION

Same sales, two stories

Axis starts at 100: “sales soar!”



Mon–Fri: 102, 103, 104, 105, 106

Axis starts at 0: nearly flat



Real change: (106 − 102) / 102 = +3.9%

Truncated axis · bar length must start at 0

Cherry-picked range · only the good months

Causation implied · ice cream vs drownings: summer drives both

TOPIC 1 · OLAP

The data cube: one structure, four moves

OLAP (Online Analytical Processing) stores data as a **cube**: **dimensions** such as store, month and product, and a **measure** such as cups sold. Like a pivot table you can turn in any direction.

Roll-up

Detail → summary: daily → monthly, city → country.

Drill-down

Summary → detail: Taiwan → Kaohsiung, Taipei.

Slice

Fix one dimension: only February.

Dice

Restrict several: fruit tea, Kaohsiung only.

WORKED EXAMPLE 4 · OLAP

A bubble tea chain: what is driving growth?

THE CUBE · CUPS SOLD

Product	Store	Jan	Feb
Milk tea	Kaohsiung	1,200	1,100
Milk tea	Taipei	1,500	1,600
Fruit tea	Kaohsiung	800	950
Fruit tea	Taipei	700	900

Roll-up to store: Taipei 4,700 · Kaohsiung 4,050

Roll-up to month: Jan 4,200 → Feb 4,550 (+8.3%)

Drill-down by product: milk tea 2,700 → 2,700 (0%); fruit tea 1,500 → 1,850 (+23.3%)

Slice, Feb only: a 2 × 2 product-by-store table

Dice, fruit tea in Kaohsiung: 800 → 950 (+18.8%)

Decision: all of February's growth came from fruit tea. Stock up on fruit and promote it, rather than both drinks equally.

TOPIC 1 · APPLICATION AND LIMITATIONS

Exploring data in practice

WHICH STATISTIC TO REPORT

Data	Report
Salaries, house prices	Median and IQR
Reaction times, no extremes	Mean and std dev
1–5 satisfaction ratings	Median or mode
Website traffic with viral days	Median; report spikes separately

LIMITATIONS

An outlier is not always an error

A flagged payment may be the fraud case you need. Investigate before deleting.

Charts involve choices

Bin width and axis range can hide or invent a pattern.

OLAP describes, not explains

It shows that fruit tea grew, not why.

How close are two data objects?

Each record becomes a vector: one number per attribute. A customer = (age, income) = (30, 500,000).

Measure	Small value	Large value
Distance	Close, alike	Far, different
Similarity	Different	Alike

EVERY DAY IT DECIDES…

which songs or products to **recommend**

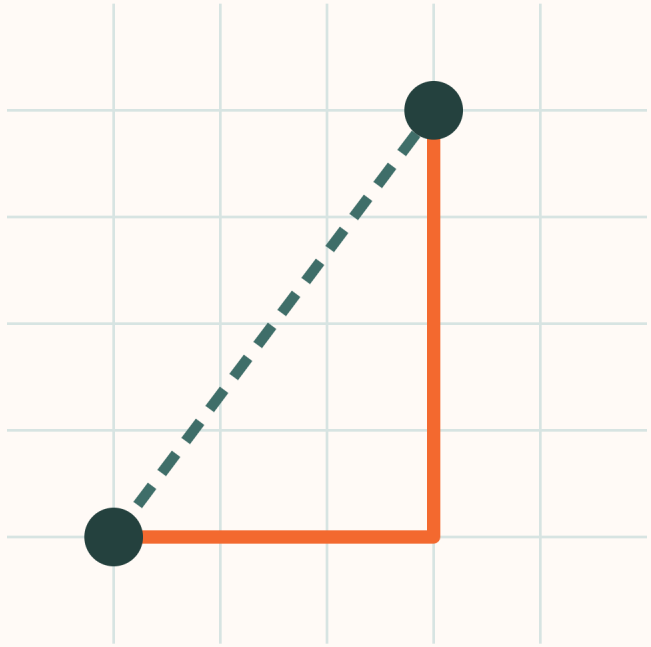
which past cases a **classifier** compares to

which customers **cluster** together

which payment is **unusual**, far from all others

WORKED EXAMPLE 5 · EUCLIDEAN VS MANHATTAN

How far does the delivery rider travel?



Restaurant at block (1, 2), customer at block (4, 6).

Euclidean (straight line): 5 blocks

$\sqrt{((4-1)^2 + (6-2)^2)} = \sqrt{(9 + 16)} = \sqrt{25} = 5$

Manhattan (along streets): 7 blocks

$|4 - 1| + |6 - 2| = 3 + 4 = 7$

A rider can't ride through buildings, so in a grid city Manhattan distance is the better estimate of travel time.

WORKED EXAMPLE 6 · THE SCALE PROBLEM

A bank looks for customers similar to Q

Customer	Age	Income (NT$)	Raw distance to Q	Scaled distance to Q
Q	30	500,000	–	–
A	50	500,000	20 (closest)	0.40
B	31	600,000	≈ 100,000	0.10 (closest)

Raw: income swamps age

A 100,000 income gap dwarfs a 20-year age gap only because income uses bigger numbers.

Scaled to 0–1 over all customers

Age 20–70, income 300k–1.3M. Q = (0.20, 0.20), A = (0.60, 0.20), B = (0.22, 0.30).

The answer flips: B, one year older with a slightly higher income, is the true look-alike.

TOPIC 2 · NORMALISATION

Two ways to put attributes on a fair scale

Min-max

$$x' = (x - \min) / (\max - \min)$$

Maps values to 0–1. Q's age: (30 − 20) / (70 − 20) = **0.20**.

Weak spot: one extreme value sets max and squashes everyone else.

z-score

$$z = (x - \mu) / \sigma$$

Standard deviations from the mean. Income 500k with μ = 700k, σ = 200k: z = **−1.0**.

Same formula as Topic 1's outlier check.

Link to Topic 1: min-max on the startup salaries gives the nine employees values from 0 to just 0.047, because the founder's 400 sets the max.

WORKED EXAMPLE 7 · JACCARD SIMILARITY

An online shop compares shopping baskets

$$J(A, B) = |A \cap B| / |A \cup B|$$

A = {rice cooker, rice, soy sauce, chopsticks}

B = {rice, soy sauce, sesame oil}

Shared: {rice, soy sauce} = 2 · All items: 5

J(A, B) = 2 / 5 = 0.40

So recommend sesame oil to A, and chopsticks to B.

Why not count matching 0s too?

The shop sells 10,000 products. Simple matching counts every product both customers skipped:

Pair	Simple matching	Jaccard
A vs B	0.9997	0.40
A vs C (phone case, charger)	0.9994	0.00

Simple matching calls both pairs near-identical. Jaccard tells them apart.

WORKED EXAMPLE 8 · COSINE SIMILARITY

Matching resumes to a data analyst job

Count how often each keyword appears:

Text	Python	SQL	Excel	Marketing
Job post	3	2	1	0
Resume A	9	6	3	0
Resume B	1	0	1	4

$$\cos(x, y) = (x \cdot y) / (\|x\| \, \|y\|)$$

Resume A is a long resume with exactly the job's skill mix (3 × the job post).

vs job	Euclidean	Cosine
Resume A	√56 = 7.48	1.00 (best)
Resume B	√24 = 4.90 (best)	4 / √252 = 0.25

Euclidean picks the marketing resume because it is short. Cosine compares direction, not length, and picks the right candidate.

WORKED EXAMPLE 9 · K-NEAREST NEIGHBOURS

Will a new loan applicant repay?

The 5 past customers closest to the applicant (scaled distance):

Customer	Distance	Outcome
C1	0.10	Defaulted
C2	0.15	Repaid
C3	0.22	Repaid
C4	0.40	Defaulted
C5	0.55	Defaulted

MAJORITY VOTE OF THE K NEAREST

k = 1: 1 Defaulted → **Default**

k = 3: 2 Repaid, 1 Defaulted → **Repay**

k = 5: 3 Defaulted, 2 Repaid → **Default**

Same applicant, three answers. k and the distance measure are decisions, so test them on held-out data.

TOPIC 2 · APPLICATION AND LIMITATIONS

Choosing a measure in practice

Data or task	Measure
Scaled sensor or numeric readings	Euclidean
City travel, robust to one big gap	Manhattan
Baskets, tags, yes/no data	Jaccard
Documents, search, embeddings	Cosine
Mixed numbers and categories	Gower, or a designed rule

LIMITATIONS

Scale first, or units decide
As the bank example showed.

Missing values
Never fill with 0; compare only available attributes.

Many dimensions
With hundreds of attributes, all distances look alike.

ANALYSIS

How the two topics connect

	Exploring data	Similarity & distance
Question	What does this column look like?	How alike are two records?
Unit of analysis	One attribute, or a group	A pair of records
Output	Statistics, charts, cube views	A distance or similarity score
Main risk	Outliers and charts that mislead	Units, sparse data, many dimensions

Exploring decides how you measure. The founder's salary that fooled the mean would also wreck min-max scaling. The z-score that flags outliers is also the standard way to scale for distance.

KEY TAKEAWAYS

Four things to remember

1 For skewed data like salaries, the median and IQR tell the truth; the mean can be gamed by one value.

2 Plot before you conclude, check where the axis starts, and drill down to find what drives a total.

3 Scale attributes before measuring distance, or the biggest units decide who is “similar”.

4 Match the measure to the data: Manhattan for streets, Jaccard for baskets, cosine for text.

SOURCES

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